

within our morning or afternoon periods). Data is first sorted in ascending order by time and trades are designated as buy-initiated or sell-initiated based on the modified Lee & Ready (1993) tic rule. Buy-initiated trades are those trades that occurred on an up tic or zero-up tic. Sell-initiated trades are those trades that occurred on a down tic or zero-down tic. Trades are not able to be designated as a buy-initiated or sell-initiated until after the first price change.

Imbalance is computed as the absolute difference between buy-initiated and sell-initiated volume for the particular period. The calculation is as follows:

$$Q = \left| \sum \text{Buy Volume} - \sum \text{Sell Volume} \right| \quad (5.4)$$

Side

The side of the imbalance is “buy” if there is more buy-initiated volume and “sell” if there is more sell-initiated volume. Mathematically, the side designation is:

$$\text{Side} = \begin{cases} +1 & \text{if } \sum \text{Buy Volume} > \sum \text{Sell Volume} \\ -1 & \text{if } \sum \text{Sell Volume} > \sum \text{Buy Volume} \end{cases} \quad (5.5)$$

Volume

Total market volume that traded over the same period used to calculate the imbalance.

$$V(t) = \sum_{i=1}^t v_i \quad (5.6)$$

Where t denotes the total number of trades during the period and v_i is the volume corresponding to the i^{th} trade in the period.

Turnover

Turnover is the total dollar value traded during the trading period.

$$\text{Turnover}(t) = \sum_{i=1}^t p_i \cdot v_i \quad (5.7)$$

where p_i is the price of the i^{th} trade.